Sampling Distribution

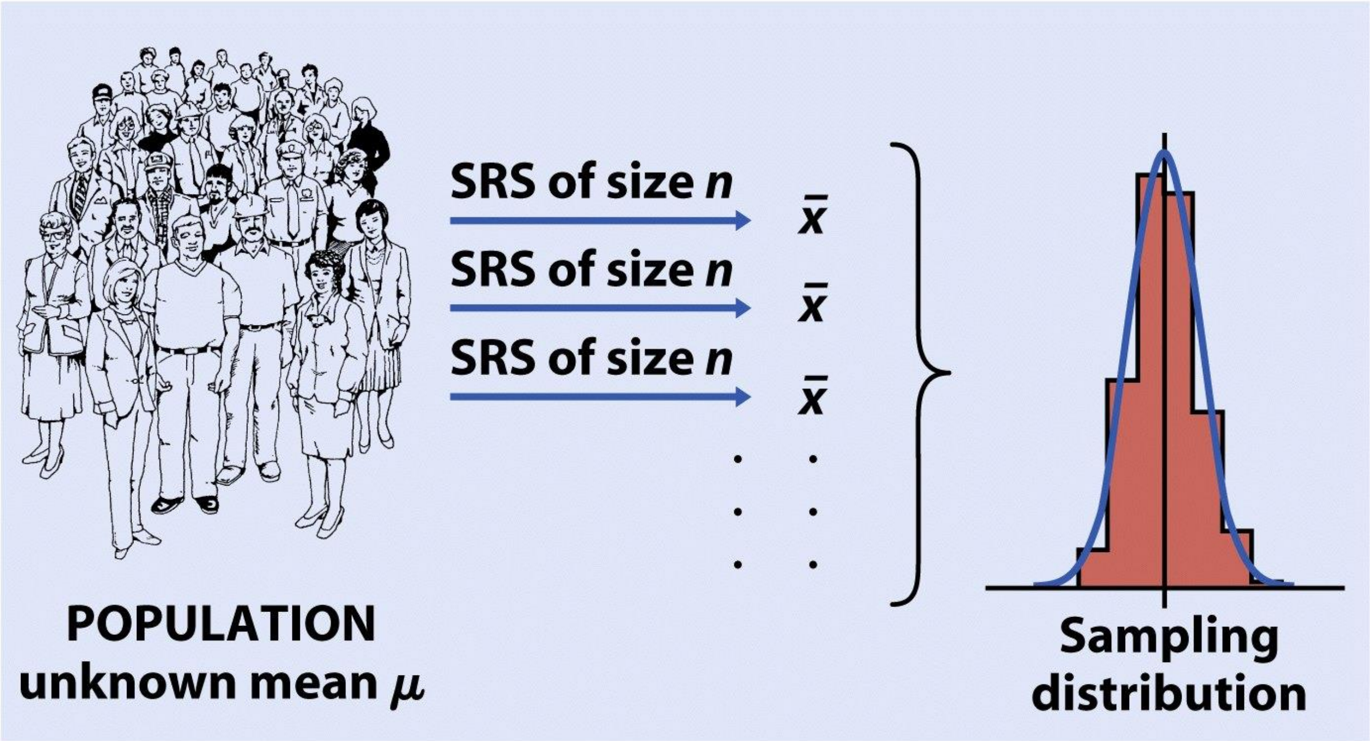

Figure 14-4a
Introduction to the Practice of Statistics, Fifth Edition

- Estimators (for example, a sample mean) are random variables
- That means they have a probability distribution: their sampling distribution

Sampling Distribution

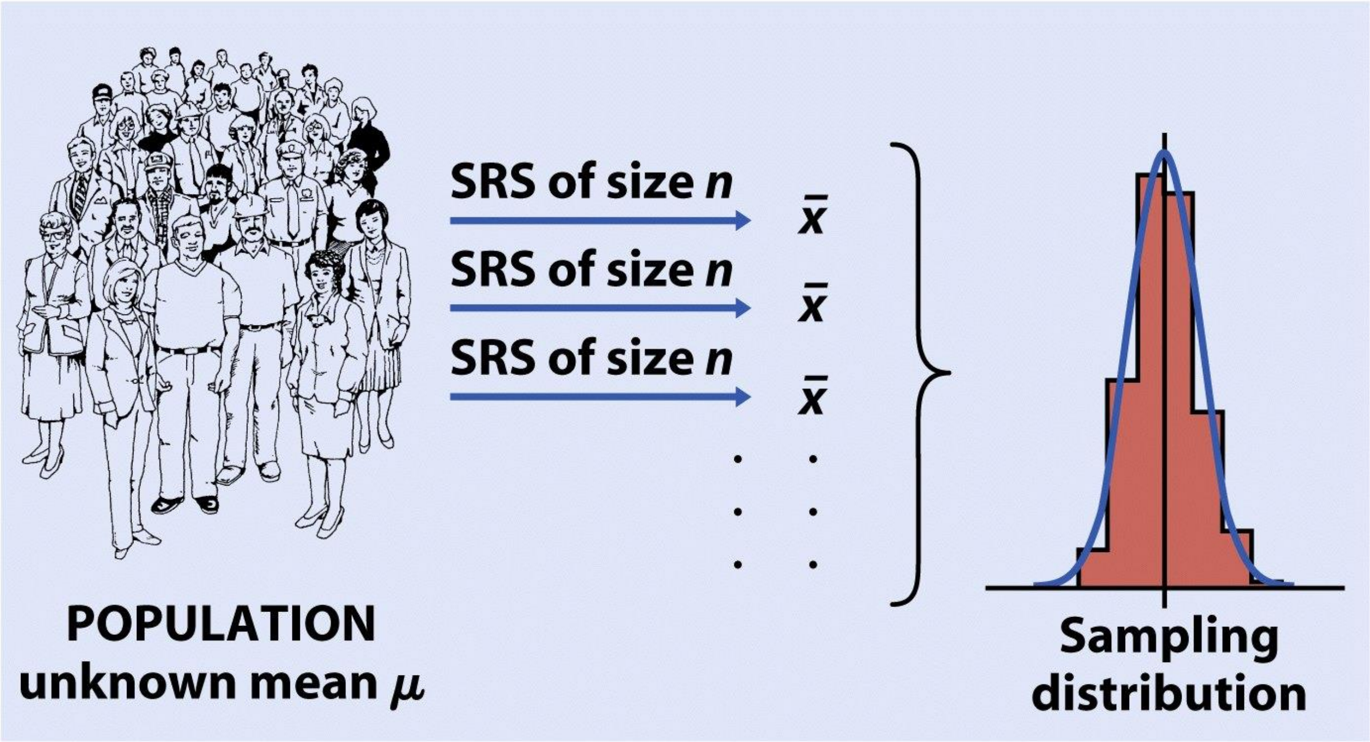

Figure 14-4a
Introduction to the Practice of Statistics, Fifth Edition

- What is the mean of this distribution?
- Bias is the difference between the mean of the sampling distribution and the true parameter

Sampling Distribution

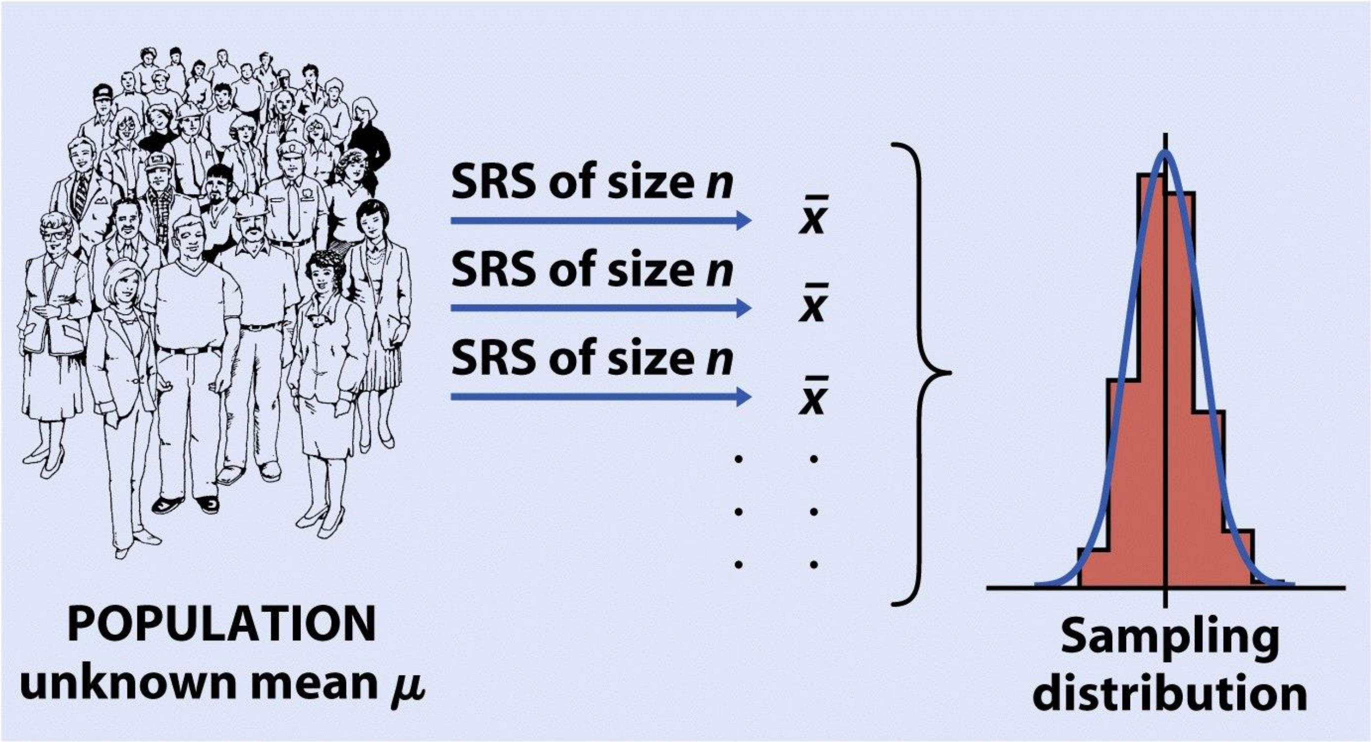

Figure 14-4a
Introduction to the Practice of Statistics, Fifth Edition

- What is the standard deviation of this distribution?
- Over repeated sampling, a 95% CI captures the true parameter 95% of the time
- p-values assess the observed statistic using the sampling distribution for that statistic <u>under the null hypothesis</u>